

Cloud Monitoring And Alert

Cloud monitoring and alerting are crucial for maintaining the health and performance of cloud-based systems. This presentation explores the key components, tools, and best practices to ensure effective monitoring and timely alerts.

Importance of Cloud Monitoring

Performance Optimization

Continuous monitoring helps identify bottlenecks in real-time, ensuring optimal performance and resource utilization across cloud applications.

Downtime Reduction

By proactively monitoring system health, organizations can detect issues early, reducing the risk of downtime and maintaining service availability.

Resource Management

Effective cloud monitoring provides insights into resource usage patterns, enabling better management of costs and resource allocation in cloud environments.

Types of Monitoring

1 Infrastructure Monitoring

This involves tracking the health of physical servers, virtual machines, and networking components to ensure system stability and performance.

2 Application Performance Monitoring

APM tools focus on the performance of applications, analyzing metrics like response times and error rates to enhance user experience.

3 Log Monitoring

Collecting and analyzing logs generated by applications and systems helps in troubleshooting issues and understanding system behavior more comprehensively.

Launch The Instance

The screenshot displays the AWS Management Console for the us-east-1 region. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security. The main content area is titled 'Instances (1/2)' and shows a table of running EC2 instances. Below the table, the details for the instance 'i-0ca16c1f2e2f234c0 (TestInstance)' are shown, including its state (Running), public IP address (3.88.178.134), and other configuration details.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

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aws | Search [Alt+S] | United States (N. Virginia) | voclabs/user3803891=tanmaydubey114@gmail.com @ 2154-7852-4846

EC2 > Instances

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive) | All states

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☒	TestInstance	i-0ca16c1f2e2f234c0	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-3-88-178-134.com...	3.88.178.134	-
☐	Bastion Host	i-010696c16da584470	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-3-239-107-123.co...	3.239.107.123	-

i-0ca16c1f2e2f234c0 (TestInstance)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID i-0ca16c1f2e2f234c0	Public IPv4 address 3.88.178.134 open address	Private IPv4 addresses 172.31.93.239
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-3-88-178-134.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-93-239.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-93-239.ec2.internal	

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To Create Alarm

The screenshot displays the AWS CloudWatch Alarms console for the us-east-1 region. The left sidebar shows the navigation menu with options like Dashboards, AI Operations, Alarms, Logs, Metrics, X-Ray traces, Events, Application Signals, Network Monitoring, and Insights. The main content area is titled 'Alarms (1/1)' and features a table with one alarm listed. The alarm is named 'CPU Alarm' and is currently in an 'Insufficient data' state. The conditions for the alarm are 'CPU Alarm > 85 for 1 datapoints within 5 minutes'. The actions are 'Actions enabled' and 'Warning'. The console also includes a search bar, filters for alarm state and type, and buttons for 'Clear selection', 'Create composite alarm', and 'Create alarm'.

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:?(selectedIds=~('CPU*20Alarm))

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CloudWatch > Alarms

CloudWatch <

Favorites and recents ▶

Dashboards

▶ AI Operations [Preview](#)

▼ Alarms [0](#) [0](#) [1](#)

In alarm

[All alarms](#)

Billing

▶ Logs [New](#)

▼ Metrics

All metrics

Explorer

Streams

▶ X-Ray traces [New](#)

▶ Events

▶ Application Signals

▶ Network Monitoring [New](#)

▶ Insights

Settings

CloudShell Feedback

Alarms (1/1)

☐ Hide Auto Scaling alarms

[Clear selection](#) [Refresh](#) [Create composite alarm](#) [Actions](#) [Create alarm](#)

Alarm state: Any Alarm type: Any Actions status: Any < 1 > ⚙

☒	Name	State	Last state update (UTC)	Conditions	Actions
☒	CPU Alarm	⏸ Insufficient data	2025-04-13 08:07:26	CPU Alarm > 85 for 1 datapoints within 5 minutes	✅ Actions enabled Warning

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Successfully Created Alarm

Sandbox Environment | Workbench - Vocareum | Alarms | CloudWatch | us-east-1

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:

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CloudWatch > Alarms

CloudWatch <

Favorites and recents ▶

Dashboards

▶ AI Operations [Preview](#)

▼ Alarms [0](#) [0](#) [0](#)

In alarm

[All alarms](#)

Billing

▶ Logs [New](#)

▼ Metrics

All metrics

Explorer

Streams

▶ X-Ray traces [New](#)

▶ Events

▶ Application Signals

▶ Network Monitoring [New](#)

▶ Insights

Settings

CloudShell Feedback

Successfully created alarm CPU Alarm. [View alarm](#)

Alarms (1) ☐ Hide Auto Scaling alarms [Clear selection](#) [Create composite alarm](#) [Actions](#) [Create alarm](#)

Alarm state: Any Alarm type: Any Actions status: Any < 1 > ⚙

☐	Name	State	Last state update (UTC)	Conditions	Actions
☐	CPU Alarm	⚠ Insufficient data	2025-04-13 08:07:26	CPU Alarm > 85 for 1 datapoints within 5 minutes	✓ Actions enabled Warning

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ENG 13:37

Add Name and Description

aws

Search

[Alt+S]

United States (N. Virginia) ▼

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CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add name and description

Step 4
Preview and create

Add name and description

Name and description

Alarm name

High CPU Alarm

Alarm description - optional [View formatting guidelines](#)

Edit Preview

This is an H1

****double asterisks will produce strong character****

This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Get The Pass Message

Sandbox Environment | Workbench - Vocareum | Launch an instance | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

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EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0ca16c1f2e2f234c0)

[Launch log](#)

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
[Manage detailed monitoring](#)

Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer
[Create Load Balancer](#)

Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a

Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.

Disaster recovery for your instances
Recover the instances you just launched into a different Availability

Monitor for suspicious runtime activities
Amazon GuardDuty enables you to continuously monitor for malicious

Get instance screenshot
Capture a screenshot from the instance and view it as an image. This is useful for troubleshooting an unreachable

Get system log
View the instance's system log to troubleshoot issues.

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Successfully Update GetMessage

The screenshot shows the AWS Lambda console for the 'getMessage' function in the 'us-east-1' region. A green notification bar at the top states 'Successfully updated the function getMessage.' Below this, there are buttons for '+ Add trigger' and '+ Add destination'. The 'Test' tab is selected, showing a red notification bar indicating 'Executing function: failed (logs [2])' with a 'Details' link and a 'Diagnose with Amazon Q' button. The 'Test event' section includes a 'Test event action' with 'Create new event' selected, an 'Event name' field containing 'testEvent1', and 'Event sharing settings' with 'Private' selected. The footer contains links for 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates.

us-east-1:console.aws.amazon.com/lambda/home?region=us-east-1#/functions/getmessage?newFunction=true&tab=testing

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Lambda > Functions > getMessage

Successfully updated the function getMessage. [X]

+ Add trigger + Add destination

Function URL [Info](#)

Code **Test** Monitor Configuration Aliases Versions

Executing function: failed ([logs \[2\]](#)) [Diagnose with Amazon Q](#)

Details

Test event [Info](#) [CloudWatch Logs Live Tail](#) [Save](#) [Test](#)

To invoke your function without saving an event, configure the JSON event, then choose Test.

Test event action

☒ Create new event ☐ Edit saved event

Event name

testEvent1

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

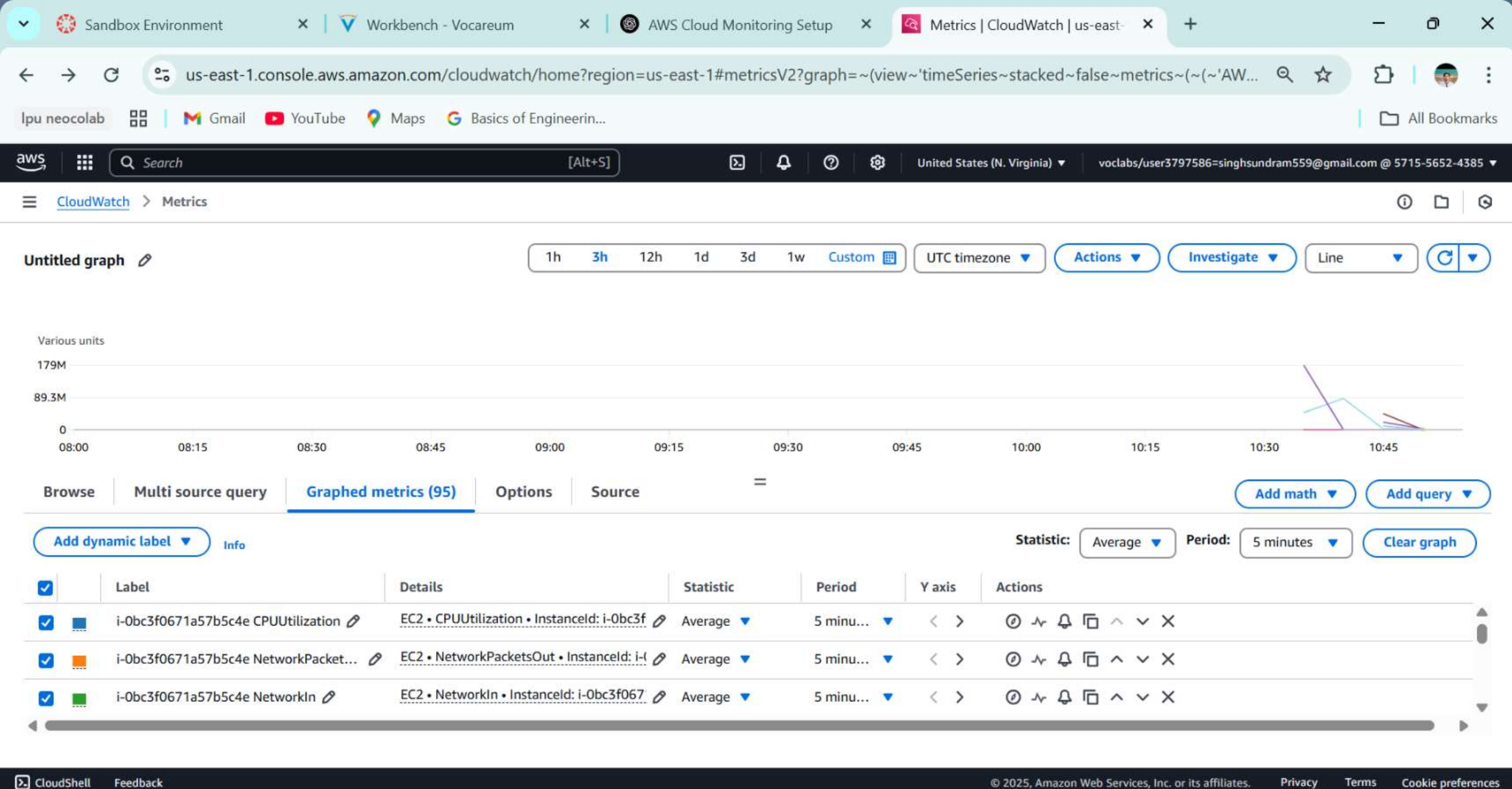
Event sharing settings

☒ Private ☐ Shareable

This event is only available in the Lambda console and to the event creator. You can configure a total of 10. [Learn more](#)

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Untitled Graph



Uploaded the GetMessage

The screenshot displays the AWS Lambda console interface for a function named 'getMessage'. The browser's address bar shows the URL: `us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/getMessage?newFunction=true&tab=code`. The AWS navigation bar at the top includes the search bar, account information, and region settings (United States (N. Virginia)).

A green notification banner at the top of the console states: "Successfully updated the function getMessage." Below this, the 'Code source' tab is active, showing the JavaScript code for the function's handler. The code is as follows:

```
1 exports.handler = async (event) => {  
2   const response = {  
3     statusCode: 200,  
4     body: JSON.stringify("Hello from Lambda!"),  
5   };  
6   return response;  
7 };  
8
```

The left sidebar contains the 'EXPLORER' panel, which shows the file structure for the 'GETMESSAGE' function, including the 'index.mjs' file. Below the explorer, there are buttons for 'Deploy (Ctrl+Shift+U)' and 'Test (Ctrl+Shift+I)'. At the bottom of the sidebar, there is a section for 'TEST EVENTS [NONE SELECTED]' with a '+ Create new test event' button.

The bottom of the console features a 'CloudShell' button, a 'Feedback' link, and copyright information: "© 2025, Amazon Web Services, Inc. or its affiliates." There are also links for 'Privacy', 'Terms', and 'Cookie preferences'.

Successfully Created GetMessage

The screenshot displays the AWS Lambda console interface for a newly created function named 'getMessage'. At the top, a green notification bar states: 'Successfully created the function getMessage. You can now change its code and configuration. To invoke your function with a test event, choose "Test".' Below this, the function name 'getMessage' is prominently displayed. To the right of the name are buttons for 'Throttle', 'Copy ARN', and 'Actions'. Under the 'Actions' dropdown, there are buttons for 'Export to Infrastructure Composer' and 'Download'. The 'Function overview' section is expanded, showing a 'Diagram' tab and a 'Template' tab. The 'Diagram' tab contains a visual representation of the function 'getMessage' with a 'Layers' section indicating '(0)' layers. Below the diagram is a '+ Add trigger' button. To the right of the diagram is a '+ Add destination' button. The 'Description' section on the right lists the following details: 'Description' (empty), 'Last modified' (1 second ago), 'Function ARN' (arn:aws:lambda:us-east-1:215478524846:function:getMessage), and 'Function URL' (empty). Below the 'Function overview' section, there are tabs for 'Code', 'Test', 'Monitor', 'Configuration', 'Aliases', and 'Versions'. The 'Code' tab is selected, showing the 'Code source' section with an 'Upload from' button. The bottom of the console features a 'CloudShell' button, a 'Feedback' link, and a copyright notice: '© 2025, Amazon Web Services, Inc. or its affiliates.' along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

getMessage | Functions | Lambda

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/getMessage?newFunction=true&tab=code

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Lambda > Functions > getMessage

Successfully created the function **getMessage**. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

getMessage Throttle Copy ARN Actions

Export to Infrastructure Composer Download

▼ Function overview Info

Diagram Template

getMessage

Layers (0)

+ Add trigger

+ Add destination

Description

-

Last modified

1 second ago

Function ARN

arn:aws:lambda:us-east-1:215478524846:function:getMessage

Function URL Info

-

Code Test Monitor Configuration Aliases Versions

Code source Info

Upload from

CloudShell Feedback

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Successfully uploaded

Upload objects - S3 bucket my- x +

us-east-1.console.aws.amazon.com/s3/upload/my-coststack-site?region=us-east-1&bucketType=general

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Upload succeeded
For more information, see the **Files and folders** table.

Upload: status Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination
s3://my-coststack-site

Succeeded
1 file, 491.0 B (100.00%)

Failed
0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 491.0 B)

Find by name

< 1 >

Name	Folder	Type	Size	Status	Error
index.html	-	text/html	491.0 B	Succeeded	-

CloudShell Feedback

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After Launch Instance

Sandbox Environment | Workbench - Vocareum | ChatGPT | Instances | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

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EC2 > Instances

EC2

- Dashboard
- EC2 Global View
- Events
- Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- Images
 - AMIs
 - AMI Catalog
- Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager
- Network & Security
 - Security Groups
 - Elastic IPs

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive) All states

Last updated 7 minutes ago

Connect Instance state Actions Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	Bastion Host	i-08413d7ce739268e2	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-54-82-108-104.co...	54.82.108.104	-
☒	Monitoring Te...	i-0a3c8f82113130ddb	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1d	ec2-54-166-1-213.com...	54.166.1.213	-

i-0a3c8f82113130ddb (Monitoring Test Instance)

10:30 10:45 11:00 11:15

Network packets out (count)

Count

350

175

0

10:30 10:45 11:00 11:15

Metadata no token (count)

No unit

1

0.5

0

10:30 10:45 11:00 11:15

CPU credit usage (count)

Count

0.412

0.206

0

10:30 10:45 11:00 11:15

CPU credit balance (count)

Count

32.5

16.3

0

10:30 10:45 11:00 11:15

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ENG 17:00

Last see the instance

Sandbox Environment

Workbench - Vocareum

ChatGPT

Instances | EC2 | us-east-1

← → ↻

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

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EC2 > Instances

🏠 ⚙️ 🗨️

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Instances (1/2) Info

Last updated 2 minutes ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☐	Bastion Host	i-08413d7ce739268e2	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-54-82-108-104.co...	54.82.108.104	-
☒	Monitoring Te...	i-0a3c8f82113130ddb	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1d	ec2-54-166-1-213.com...	54.166.1.213	-

i-0a3c8f82113130ddb (Monitoring Test Instance)

Alarm recommendations

Investigate with AI - new

1h 3h 12h 1d 3d 1w Custom

UTC timezone

Explore related

CPU utilization (%)

Percent

4.95

2.47

0

10:30 10:45 11:00 11:15

Network in (bytes)

Bytes

9.39M

4.69M

0

10:30 10:45 11:00 11:15

Network out (bytes)

Bytes

26.0k

13.0k

0

10:30 10:45 11:00 11:15

Network packets in (count)

Count

1.17k

586

0

10:30 10:45 11:00 11:15

Network packets out (count)

Count

0

0

0

10:30 10:45 11:00 11:15

Metadata no token (count)

Count

0

0

0

10:30 10:45 11:00 11:15

CPU credit usage (count)

Count

0

0

0

10:30 10:45 11:00 11:15

CPU credit balance (count)

Count

0

0

0

10:30 10:45 11:00 11:15

CloudShell

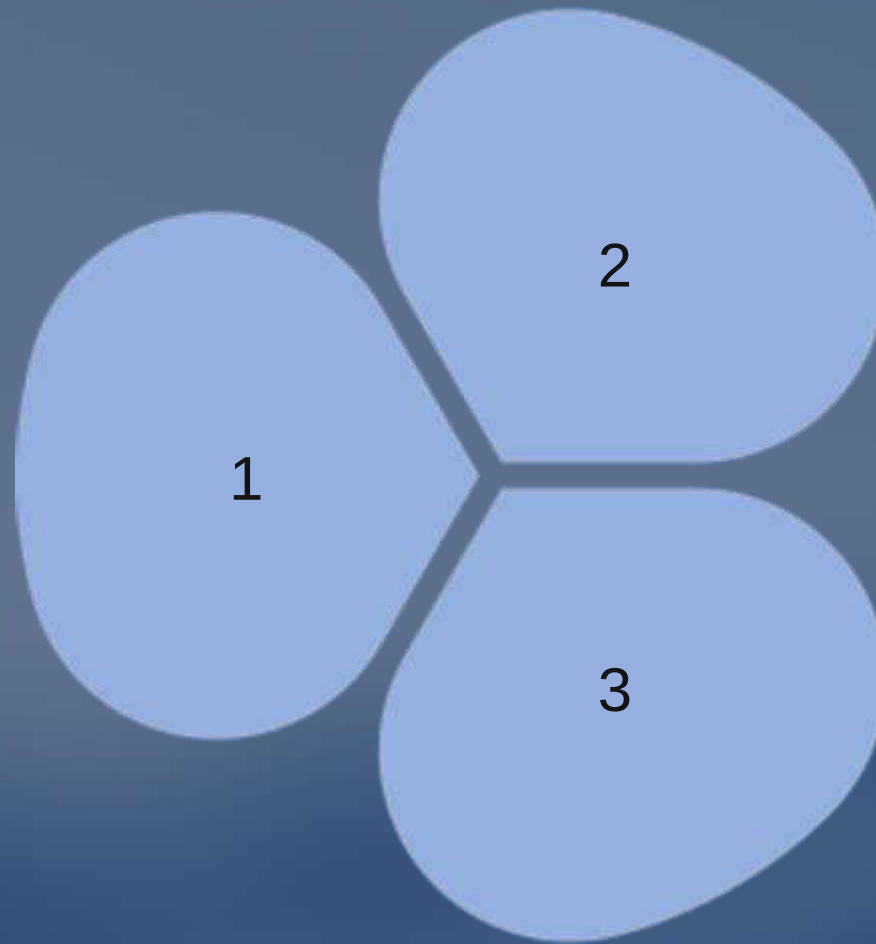
Feedback

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Key Monitoring Tools

Prometheus

An open-source monitoring system that collects and stores metrics as time series data, providing robust querying and alerting capabilities.



Grafana

A visualization tool that helps create dashboards to visualize metrics from various data sources.

Cloud Provider Services

Major cloud providers offer built-in monitoring solutions such as AWS CloudWatch and Azure Monitor.

Alerting Strategies

Threshold-based Alerts

Setting predefined thresholds for metrics to trigger alerts helps teams respond promptly to common performance issues, minimizing manual monitoring efforts.

Anomaly Detection

Utilizing machine learning algorithms, anomaly detection alerts can automatically signal unusual patterns in behavior that may indicate underlying issues, enhancing proactive monitoring.

Escalation Policies

Establishing clear escalation policies ensures that alerts are directed to the appropriate team members and acted upon swiftly, reducing potential impacts on service and performance.

Best Practices for Monitoring

Establish Clear Metrics

Identify key performance indicators (KPIs) that align with business objectives to measure and manage cloud performance effectively.



Regular Review and Adjustments

Continuously reviewing monitoring setups and alerts can help maintain their effectiveness, adapting to changing application loads and business needs.

Automate Responses

Where possible, automate responses to specific alerts using scripts or cloud automation tools, reducing manual efforts and response times.

Challenges in Cloud Monitoring



Complexity of Hybrid Environments

Monitoring hybrid cloud environments demands comprehensive strategies and tools for visibility.



Data Overload

The vast amount of data can lead to alert fatigue if not filtered and prioritized.



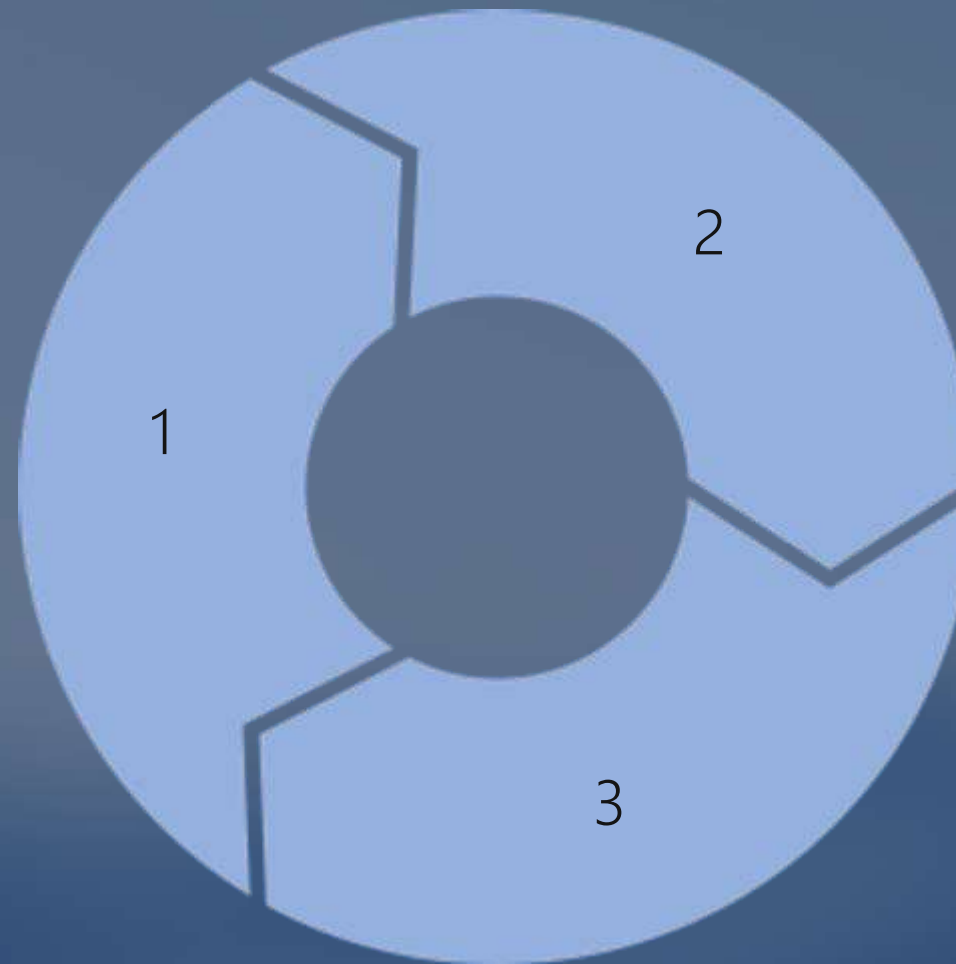
Integration with Existing Tools

Monitoring solutions must integrate seamlessly with existing workflows for effective incident management.

Future Trends in Monitoring

AI and Machine Learning

The growth of AI in monitoring tools enhances predictive abilities and minimizes false alerts.



Serverless Monitoring

Tools developed for serverless architectures will track transient metrics effectively.

Enhanced User Experience Focus

Monitoring solutions will prioritize insights into end-user interactions across channels.